

Emiliano Garcia

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MACHINE LEARNING ENGINEER | APPLIED AI SYSTEMS | EIT

Applied ML engineer and graduate researcher building AI/ML systems end to end, from preprocessing and training pipelines to evaluation, deployment-ready services, and production tooling. Experience spans multimodal modeling, GenAI systems, RAG, LLM adaptation, model evaluation, and ML tooling at Autodesk Research, with a focus on reproducible systems that solve real problems.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, C++

ML: PyTorch, TensorFlow/Keras, scikit-learn, Transformers; multimodal ML, CNNs, deep learning

Experimentation: ablations, cross-validation, hyperparameter tuning, robustness testing, explainability

Systems: Git, Docker, CI/CD; Slurm/sbatch (HPC), multi-GPU (DataParallel); FastAPI, Node.js; AWS (EC2/S3/IAM), GCP

Data: NumPy, Pandas; preprocessing pipelines, embeddings, spectrograms, audio processing

SELECTED PROJECTS

RAG Benchmark Service

Benchmark-driven retrieval-augmented generation system for scientific QA

- Built an evaluation-first retrieval system on SciFact (5,183 docs; 300 labeled queries), implementing Recall@K, nDCG@K, and MRR@K with timestamped comparison reports.
- Implemented BM25 and dense/FAISS retrievers behind one interface; benchmarked BM25 (R@10 0.776, nDCG@10 0.652) vs dense (R@10 0.783, nDCG@10 0.645) to quantify trade-offs.
- Deployed a FastAPI service with JSONL logging and /metrics; added PSI drift checks, weekly eval, gated promotion, and 239 tests.

PocketGuide

Domain-adapted LLM for structured travel guidance with offline inference

- Built an evaluation-first LLM adaptation workflow with a fixed 20-prompt benchmark suite (parse success, schema compliance, uncertainty markers) to measure improvements before deployment.
- Implemented a teacher-student synthetic instruction pipeline with multi-stage quality gating and provenance tracking (prompt hashes, config snapshots) to generate high-fidelity training data.
- Fine-tuned Llama-2-7B with LoRA across 5 iterations, improving parse success from 80% to 100%; quantized to GGUF (llama.cpp) for offline inference on consumer hardware.

EXPERIENCE

Wilfrid Laurier University, Waterloo, ON

Sep 2025 to Present

Graduate Research and Teaching Assistant

- Developing deep learning pipelines for multimodal cognitive health assessment (speech, text, and auxiliary features).
- Built preprocessing workflows (feature extraction, spectrograms, embeddings) and evaluated frozen-encoder audio/text models for low-data cognitive assessment.
- Ran subject-independent 5-fold CV with ablations and robustness checks; late-fusion reached 0.856 ± 0.038 macro-F1 and 0.922 ± 0.030 AUC; first-author paper accepted at IEEE EMBC 2026.
- Supporting course instruction through grading, feedback, and student guidance in ML, data, and software engineering.

Autodesk Research (Autodesk), Toronto, ON

May 2023 to Sep 2023

Software Developer Intern

- Built a schedule-optimization algorithm from scratch for HCBD; evaluated millions of candidate scenarios and achieved ~60% lower walking distance than manually constructed baselines.
- Prototyped an ML-based approach to learn occupancy patterns and guide layout/scheduling decisions; validated performance using objective walk-distance metrics across large-scale scenario sweeps.
- Engineered data processing pipelines to transform raw architectural geometry into analyzable datasets/graphs; implemented path modeling to simulate occupant movement and quantify walk-distance outcomes.
- Developed geometry processing and visualization utilities to communicate findings; presented results and documentation to researchers and cross-functional stakeholders.

Thompson Rivers University, Kamloops, BC

May 2022 to Jan 2023

Undergraduate Research Assistant

- Conducted ML research with Dr. Emad Mohammed on cognitive similarities between COVID-19 and Alzheimer's disease using audio processing and statistical learning.
- Built automated pipelines to transform speech into 2D frequency representations for computer-vision-based detection.
- Trained and evaluated ensemble CNN approaches for Alzheimer's detection from speech; performed clustering on demographic and cognitive features.
- Contributed to EMBC 2026 accepted paper through analysis, methodology documentation, and result visualization.

Thompson Rivers University (Enterprise Systems), Kamloops, BC

Sep 2023 to Dec 2023

Software Analyst Intern

- Built reporting, data extraction, and transformation automations using FormFusion/Argos, SQL, and Oracle databases to streamline institutional workflows.

EDUCATION

Master of Applied Computing (M.A.C.)

Sep 2025 to Present

Wilfrid Laurier University, Waterloo, ON.

- Thesis stream; graduate research on multimodal ML (speech + text) under limited-data constraints.
- Graduate Research Assistant / Teaching Assistant.

Bachelor of Engineering in Software Engineering

Sep 2020 to Apr 2025

Thompson Rivers University, Kamloops, BC.

- Co-op; Undergraduate Research Assistant in machine learning for speech analytics.

RESEARCH & PUBLICATIONS

Kolmogorov-Arnold Networks for Alzheimer's Disease Detection using Multimodal Speech Analysis

First Author, IEEE EMBC 2026.

- Frozen-encoder multimodal AD detection system reached 0.856 ± 0.038 macro-F1 and 0.922 ± 0.030 ROC-AUC with learned late fusion.

Analyzing COVID-19 Cough Sounds and Alzheimer's Speech with Deep Learning

Co-author, IEEE EMBC 2026.

- Shared 2D audio-analysis framework found representation-level proximity between AD speech and COVID coughs; weighted CNN ensemble reached 73% accuracy.

Awards: Laurier Graduate Scholarship; URA Grant; Jorge Campos Award; Dean's List.

LEADERSHIP

Lululemon, Guest Experience Lead / Educator

Apr 2024 to Present

- Coached team members, supported operations, and improved consistency through structured feedback.